Discover Food

https://doi.org/10.1007/s44187-026-00806-2

Article in Press

A Documentation on the traditional fermentation practices and contemporary perspectives of *Mishti doi* in Bengal

Received: 24 June 2025

Accepted: 6 January 2026

Published online: 21 January 2026

Cite this article as: Chakraborty P. & Ghatani K. A Documentation on the traditional fermentation practices and contemporary perspectives of *Mishti doi* in Bengal. *Discov Food* (2026). https://doi.org/10.1007/s44187-026-00806-2

Priya Chakraborty & Kriti Ghatani

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

ARTICLE IN PRESS

A documentation on the traditional fermentation practices and contemporary perspectives of *Mishti doi* in Bengal

Priya Chakraborty[1] and Kriti Ghatani [1]*

[1]Food Microbiology laboratory, Department of Food Technology, University of North Bengal, Raja Rammohunpur, Siliguri, District-Darjeeling, 734013, West Bengal, India

Priya Chakraborty: Research Scholar, [1]Food Microbiology laboratory, Department of Food Technology, University of North Bengal, Raja Rammohunpur, Siliguri, District-Darjeeling, 734013, West Bengal, India, rs_priya@nbu.ac.in

*Corresponding author: *Dr. Kriti Ghatani, Assistant Professor, [1]Food Microbiology laboratory, Department of Food Technology, University of North Bengal, Raja Rammohunpur, Siliguri, District-Darjeeling, 734013, West Bengal, India, ghatanik@nbu.ac.in

ARTICLE IN PRESS

ARTICLE IN PRESS

Abstract:

This study aims to document a popular traditional fermented milk product, *Mishti doi*, from North Bengal, India. *Mishti doi* is a popular regional delicacy known for its sweet, caramelised flavour, thick and creamy consistency. The consumption of *Mishti doi* in Bengali culture is deeply rooted in tradition, with applications ranging from everyday meals to religious rituals and modern culinary innovations. The study employed a descriptive and field based analytical approach with observations and responses being thematically analysed to identify common trends in fermentation steps, ingredients and product variations. To capture authentic local practices, a total 40 participants were interviewed face to face and asked open ended questions. Analysis was done through manual coding and thematic grouping of responses to understand cultural significance, traditional preparation methods and challenges in local dairy practices. Through interactions with traditional sweet makers, the documentation highlights the craftsmanship involved in making *Mishti doi*, demonstrating the techniques passed down through generations. It also discusses the difficulties of preserving authenticity in the face of industrialisation and commercialisation. Additionally, the survey analyses the influence of modern food trends on this age-old delicacy. The findings contribute to the preservation and promotion of *Mishti doi* as an integral part of Bengal's cultural heritage while exploring opportunities for its broader market potential.

ARTICLE IN PRESS

Keywords: *Mishti doi*, West Bengal, Fermented Food, Probiotics, Ethnicity

Introduction

The West Bengal state is a part of eastern India that shares an international border with Bangladesh as well as Nepal and Bhutan. It extends from the Himalayas in the north to the Bay of Bengal in the south. West Bengal is surrounded by Assam and Sikkim in north east, Orissa in the southwest, and Jharkhand and Bihar in the west. [1]. The Ganga river has traditionally divided Bengal into South Bengal and North Bengal, which are further divided into Terai and Dooars regions. The northern part of West Bengal is known as North Bengal. It is the west of the Jamuna river and north of the Padma river, which includes the Barind Tract. The North Bengal section includes the Jalpaiguri division (Alipurduar, Cooch Behar, Darjeeling, Jalpaiguri, and Kalimpong) as well as the Malda division (Uttar Dinajpur, Dakshin Dinajpur, and Malda) [2] (Figure 1). West Bengal is known for its rich biodiversity, featuring a wide variety of flora and fauna

ARTICLE IN PRESS

due to its climatic variations. Bengal has a diverse population, including Bengalis, tribal groups, and smaller ethnic minorities. The Bengali community represents the largest population group, while tribal communities such as the Santhal, Munda, Oraon, Koch-Rajbongshi, and Toto also form an integral part of the state's demographic composition. Different food products are also available due to differences in culture, geographical indication, and raw material availability (grains, milk, plant sources) [3]. West Bengal is renowned for its rich and diverse culinary heritage, where traditional dishes featuring fish, rice, and an array of sweets hold a prominent place. A variety of fermented foods are also produced, which can be divided into four classes (cereal- and pulse-based, plant-based, dairy-based and fish-based) [4]. Among of them fermented milk products are most common. Local people prepare traditional fermented foods by using ancestral ethnic knowledge. Some examples of naturally fermented milk products include *lassi*, *mishti doi*, *rabri*, *shrikhand*, *chhu* and *chhurpi* [5]. Among them *Mishti doi* is one of the popular dessert throughout the Bengal. *Mishti doi* has been for centuries an important part of Bengali cuisine, with preparation methods passed down through generations, especially in the regions of northern districts. These traditional techniques, such as prolonged boiling of milk, caramelisation of sugar or jaggery and fermentation in earthen pots are passed down through experience rather than written instructions. This result in rich regional diversity in flavour, texture, and production style. Despite the cultural and economic importance of *Mishti doi*, there is a noticeable lack of systematic documentation describing how it is traditionally made in real production settings, particularly in North Bengal, where these practices are deeply rooted. Existing literature mainly focusses on another fermented milk products or commercially produced dahi, providing only broad descriptions of fermentation processes that fail to capture the distinct, artisanal steps taken by rural and small scale *Mishti doi* artisans. As a result, several significant factors remain unexplored, forming a clear research gap. There is limited knowledge of the exact preparation sequence employed by traditional producers, the specific heating and caramelisation techniques and their effects on product characteristics are poorly understood. The nature and the composition of the indigenous starter cultures used by communities remain undocumented and the influence of utensils, particularly earthen pots, on fermentation dynamics is largely unexplored. Furthermore, there is little information available about local hygiene perceptions, safety practices, and the environmental factors that influence fermentation outcomes. Because these methods have existed for decades without formal records and are now being challenged by modernisation, extensive field based documentation is critical for preserving traditional knowledge and understanding how these long practiced methods influence sensory quality and safety. Most previous studies only mention general fermentation methods and do not explain the specific steps, tools and local practices

ARTICLE IN PRESS

ARTICLE IN PRESS

used in authentic *Mishti doi* production, particularly in North Bengal. This study employs a field survey and on site observations of *Mishti doi* producers to provide a detailed, evidence based description of how *Mishti doi* is traditionally prepared, cultural significance and also identifying the gaps in research and suggesting potential areas for future exploration.

Traditional milk products of India

People use milk as part of regular diet in India, where the majority of citizens are vegan. Cattle were an important part of Vedic traditions prior to 800 BCE. Literature refers that, milk (cow, buffalo, and goat) was consumed in raw form [6]. One of the simplest technique for protecting milk nutrients and increase the products shelf life for human consumption is fermentation. The majority of natural milk products are fermented either spontaneously or by the back-sloping method. Back-slopping involves inoculating a fresh batch of a fermented product with a portion of a previous batch of the fermented food. Fermentation is an enzyme catalysed, metabolic process, where micro-organisms convert the complex carbohydrate molecules to smaller components like alcohol and acid anaerobically. The process of fermentation not only expands the product's life span but also enhances flavour and makes milk easier to digest. The most widely consumed and ancient type of fermented milk in India, is *dahi* that produced and consumed in every family [3]. Indian fermented milk products are distinctive because of dynamic sociocultural practises that have evolved over several years. Some of the naturally fermented milk products of India include *lassi, mishti dahi, rabri, shrikhand, chhu and chhurpi*, [5, 7].These products enormous microbial diversity and distinctive metabolic footprints match their regionally specialised technological, economic, and sensory characteristics [8].

History of *Mishti doi*

Mishti doi is a common fermented milk product which is very well in eastern India, particularly in West Bengal, Bihar, and Assam. *Mishti doi i*s originated in the Bogra district of Bangladesh. Regardless of cast, creed or culture *Mishti doi*, is an essential part of every Bengali's kitchen. Indo-Aryan ethnoliguistic group are known as bangalees or Bengalis and they are historically connected to the Bengal region of South Asia. They belong to the eastern group of Indo-Aryan languages and correspond to the Magadhan subfamily [9]. In both rural and urban Bengal, *Mishti doi* is regarded as a special dessert for ceremonial events. The typical brown hue, firm texture, creaminess, and caramelised flavour of good *Mishti doi* are all there. [9]. It is currently commercially made and distributed throughout the nation Figure 3. The product is commonly sold in various-sized earthen pots. *Streptococcus salivarius subsp. thermophilus*, *Lactobacillus acidophilus, Lb. delbrueckii subsp. Bulgaricus* and *Lactococcus lactis subsp. lactis* are the predominant microbes found in *Mishti doi* [10, 11, 12, 13]. These microorganism ferment lactose into lactic

ARTICLE IN PRESS

acid. This acid contributes to the overall flavour of the product and subsequently improves preservation and also developed its unique distinctive body texture. *Enterococcus italicus*, *Lactobacillus helveticus*, *Lactococcus raffinolactis* , *Bifidobacterium mongoliense*, *Hafnia alvei*, *Lactococcus lactis*, *Acetobacter okinawensis*, *Streptococcus thermophilus*, *Thermus thermophilus*, *Leuconostoc citreum*, *Leuconostoc pseudomesenteroides*, *Acetobacter orientalis*, *Lactobacillus gallinarum*, *Lactococcus chungangensis* are the predominant bacteria found in *Mishti doi* as per the metagenomics study [14].

Methodology

The survey was a descriptive method of research and relied mainly on field observations and open-ended questions and face-to-face interviews was done to collect both qualitative and quantitative data about *Mishti doi* preparation methods, components, preservation, sensory desires, and cultural significance. In addition to the survey, a thorough analysis of literature was conducted, gathering critical information from various sources, such as books, journal articles, and other published materials. Sources were collected from academic databases which are commonly used for research, including Google Scholar, Web of Science, Springer, ScienceDirect, and Scopus. The review focused on publications from 1990 to 2024, with the scholarly papers analysed using a mix of qualitative and quantitative methodologies.

ARTICLE IN PRESS

Study area

The survey was carried out in various rural and urban areas of North Bengal (represented in Figure 1), India. The North Bengal section includes the Jalpaiguri division (Alipurduar, Cooch Behar, Darjeeling, Jalpaiguri, and Kalimpong) as well as the Malda division (Uttar Dinajpur, Dakshin Dinajpur, and Malda). Information gathered from local residents during the survey is crucial for preserving traditional knowledge related to *Mishti doi* production.

Study design

Instead of rather using fixed multiple-choice questions, an open-ended questionnaire was created to allow participants to freely express their true thoughts and experiences in their own words. The study was carried out in various part of North Bengal, where local people frequently purchase and consume *Mishti doi*. This direct conversation helped to create an atmosphere in which participants could speak freely, resulting in deeper and more significant responses. Respondents are allowing to express their opinions in detail, such as what they like best about *Misti doi* preparation, what improvements they expect, how frequently they buy it, and what impacts their products.

Study population

ARTICLE IN PRESS

Overall population comprises households, small-scale producers (e.g., homemakers, milk vendors) and adult consumers (≥18 years) who regularly consume *Mishti doi*. Inclusion criteria include the participants must be residents of the study area, aged 18 or older, eager to provide informed consent, and participant in the preparation, sale, or consumption of *Mishti doi*. Exclusion criteria include temporary visitors and individuals who are unable to provide informed consent.

Sample size and sampling strategy

A total of 40 individuals over the age of 18 were approached in the surrounding areas. Each of the interviews was conducted personally. A multistage, mixed sampling approach was used. In the first stage, eight districts were chosen at random from the study region to ensure broad regional coverage. In the second stage, rather than formally selecting blocks or wards, renowned *Mishti doi* producing areas within every district were chosen by using local information. Local residents, shopkeepers were consulted to identify locations known for traditional *Mishti doi* preparation. These reputation-based sites where *Mishti doi* is well known, frequently prepared, or commercially sold were then specifically incorporated into the survey. Within each chosen region, households, small-scale producers and sweet shops engaged in *Mishti doi* preparation were approached. When there was no formal list of producer available, snowball sampling was used, in which initial respondents recommended to other producers or vendors. This approach helped the survey to gather traditional practices and regionally famous methods that might not be recorded in official sampling frameworks.

ARTICLE IN PRESS

Data collection tools and procedures

The survey was conducted using paper and pen with closed and open-ended questions collected socio-demographic information, consumption frequency, ingredients, fermentation method, heating or sterilisation steps, flavouring practices, storage conditions, and opinions on safety. The photographs of the preparation sites, utensils, and fermentation steps were taken only with the participant explicit permission, ensuring that all images were taken respectfully and solely for documentation purposes. A semi-structured interview was used with producers to obtain information about the history of preparation, recipe variations, production challenges, and commercialisation opportunities. Direct observation was used during on-site visits to document hygiene practices, types of utensil, incubation time, heating temperatures and fuel sources.

Variables and measures

ARTICLE IN PRESS

Survey was mainly focused on qualitative understanding of the traditional *Mishti doi* making process. Traditional methods of transmission, cultural significance, production scaling challenges, and safety perceptions are among the qualitative themes explored. Quantitative factors include consumption frequency, milk type (buffalo/cow/mixed), heating method, fermentation starter (curd/commercial starter culture/spontaneous), incubation temperature/time, sweetening agent, storage practice, and willingness to buy commercial *Mishti doi*. Secondary variables include age, gender, education level, occupation, household size, cost, and perceived health consequences.

Ethical approval and participant consent

All participants provided verbal informed consent prior to any discussions, observations, or photographs. Participation was entirely voluntary and respondents had the option to respond to any question or withdraw at any time without consequence. Photographs were taken only with participant's explicit permission and they were assured that the images would be used solely for academic documentation and not shared publicly. To maintain confidentiality and respect the privacy of study participants, all personal information was anonymised and data were securely stored. This survey did not involve any kind of clinical procedure or risk to the respondents of North Bengal region. Hence, institutional ethics committee of the University of North Bengal waived the need for approval.

ARTICLE IN PRESS

Limitations

Due to the informal framework of traditional *Mishti doi* production, popular locations for production were identified through neighbourhood support and snowball sampling was used. As a cross-sectional qualitative study, data only include practices observed and reported during the survey period.

Results and Discussions

The West Bengal state's precise coordinates are 27°13'15" and 21°25'24" north latitudes, and 85°48'20" and 89°53'04" east longitudes. The North Bengal section includes the Jalpaiguri division (Alipurduar, Cooch Behar, Darjeeling, Jalpaiguri, and Kalimpong) as well as the Malda division (Uttar Dinajpur, Dakshin Dinajpur, and Malda). The survey adequately included eight districts of the study region (presented in figure 1). Within these districts, a total of 40 peoples were identified based on recommendations from local residents, vendors, and sweet shop. These areas were well-known for their traditional fermented milk practices, with numerous small scale producers or sweet shops preparing *Mishti doi* for the past decade. The selected sites represented a wide range of environments, from busy market-based sweet shops to household level, allowing the study to know differences in preparation styles, equipment used, and cultural

significance across districts. The survey was focused on regional hubs of *Mishti doi* production, which represent areas where traditional practices are preserved. Manual coding of survey responses allowed similar statements to be clustered, and thematic grouping revealed major patterns in traditional *Mishti doi* preparation. These themes included differences in heating practices, starter culture handling, common storage issues and producers perceptions of product safety and shelf life. The grouped themes helped to clarify both respondents shared practices and the localised variations seen across preparation sites.

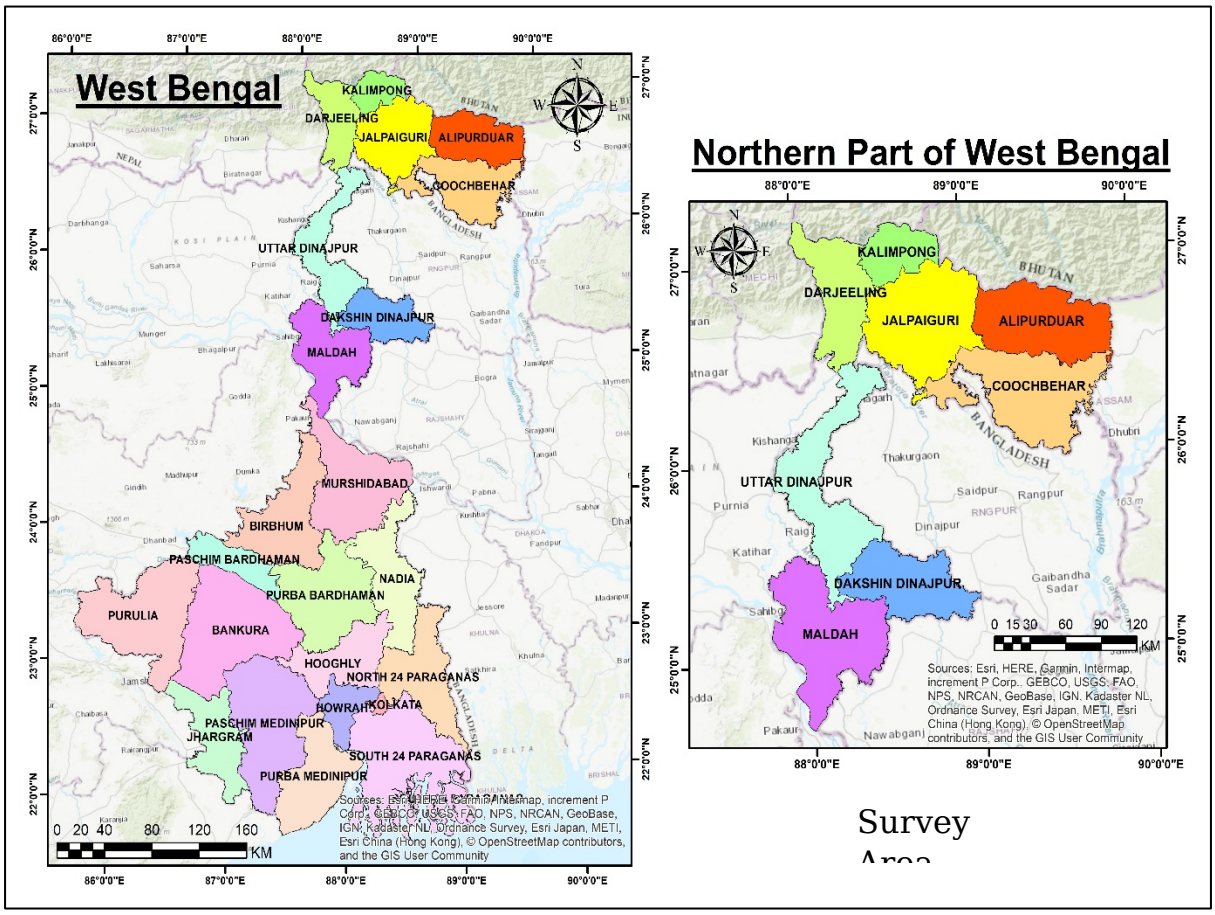


Figure 1: Geographical map of West Bengal, India

Socio demographic characteristics of participants

The survey results revealed that a significant portion of households of the West Bengal rely on agriculture and related activities for a living. A large number of respondents reported being involved in crop cultivation, while many families also engaged in animal husbandry, small-scale dairy rearing, and livestock management. These activities have been identified as important sources of income for small and marginal farmers, landless labourers, and rural women who actively participate in milk production

ARTICLE IN PRESS

and daily livestock care. Participants also stated that dairy-related activities, such as milk collection, curd setting, and small-scale *Mishti doi* preparation, help to improve household nutrition and local food availability. The livestock sector constitutes a vital part of the agricultural economy. Livestock production and crop farming are deeply interconnected, each contributing essential resources to the other. This integration supports key livelihood activities, enhances household nutrition and health, and provides additional income and employment opportunities. In rural areas of West Bengal, two primary grazing systems were identified: controlled grazing and free (or uncontrolled) grazing. Agricultural land is commonly used for grazing. Additionally, animals are often tethered along riverbanks, near ponds and canals, beside roads, and in open fields. Uncontrolled grazing practices were closely linked to the types of crops grown and the cropping intensity in a given area. In the absence of free grazing land, villagers resorted to using crops like vegetables, oilseeds, paddy, and pulses as livestock feed. An analysis of feed selection revealed that the preference for certain feed types varied across livestock species and was strongly influenced by their species-specific characteristics. Common indigenous breeds include Sahiwal, Jersey crossbreeds, and Holstein Friesian crossbreeds. The state annually produces more than 6 million tonnes of milk according to recent data. Women in farming households played a key role in managing dairy activities. They were primarily responsible for tasks such as cleaning animal shelters and feeding the livestock. Men generally took charge of decisions related to purchasing and selling animals, as well as transporting milk to sweet shops, markets, and homes. Milk was collected from farms and delivered to local collection points, often managed by village communities or cooperatives. Here, milk was weighed and its quality was tested. Based on both quantity and quality, farmers receive their payments. After collection, the milk is transported to processing plants where it undergoes pasteurization to ensure safety and homogenization for consistency and distributed to nearby sweet makers or wholesalers. Large-scale sweet manufacturers often purchase dairy supplies directly from cooperatives or processing units to ensure reliable quality and supply. Milk was turned into other dairy products such as *ghee*, *paneer*, *khoya*, or *chhena*. Using these ingredients, sweet makers craft traditional Bengali delicacies like *rasgulla*, *sandesh*, *chamcham* and different fermented milk product like *mishti doi*, *rabri*, *lassi* etc. Ethnic peoples using unique techniques for preparing these fermented milk product and it remain unchanged despite advancements in Indian cuisine. The preparation of these products is unique to Bengali community and cannot be found anywhere else. Local people using cow or buffalo milk for preparation of *Mishti doi*. Ethnic people used trial and error approaches to improve food quality. Indigenous knowledge is passed down and modernised with each generation. Eventually, the finished sweets are sold in sweet shops, local marketplaces, and increasingly through online platforms. They are especially popular during cultural celebrations, festivals, and are also exported. People in West Bengal region regularly

ARTICLE IN PRESS

ARTICLE IN PRESS

consume *dahi* and it is traditionally offered to God during worship. The product's approximate selling price is Rs. 150-250 per kg.

Traditional Process of *Mishti doi* Preparation

Following data collection, responses were carefully reviewed, grouped into subdivisions and interpreted to better understand about *Mishti doi* preparation. During the analysis, special attention was given to the different regional variations of *Mishti doi* mentioned by the respondents.

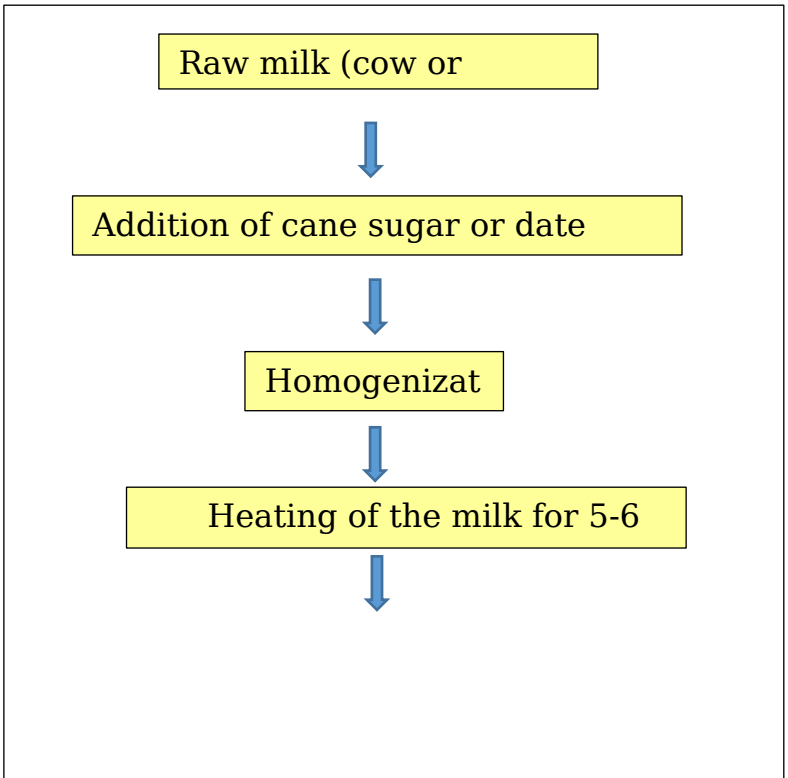
Ingredients and core preparation process:

The survey results revealed significant consistency in the core ingredients used in *Mishti doi* preparation, which included full-fat milk, sugar or caramelised sugar and starter culture. Milk source and fat content varied while cow's milk was the most common, some producers used a combination of cow and buffalo milk to improve texture. Heating was an important step, with milk typically boiled and reduced by 10-40% to develop the distinctive caramelised sweetness. The exact amount of reduction differed depending on producer preference, heating equipment, and desired sensory characteristics.

Heating, sterilisation and fuel source:

Milk was heated for an extended period of time in order to develop flavour as well as for the sterilisation purpose. Some producers sterilised their utensils before use by boiling or sun-drying them, while the maximum people used simple rinsing methods. Fuel sources included LPG (the most common), firewood. Producers who used firewood reported temperature inconsistencies caused by fluctuating flame intensity, affecting milk reduction uniformity and final texture. Due to slow heat exposure contribute to richer colour and flavour.

ARTICLE IN PRESS



ARTICLE IN PRESS

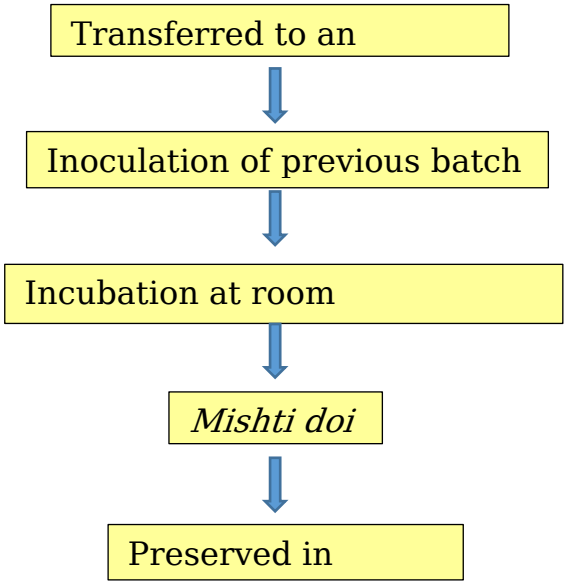


Figure 2: Preparation of *Mishti doi*

ARTICLE IN PRESS

Fermentation practice:

Fermentation methods differed greatly across sites. Clay pots (matir bhar) remained the most popular vessels due to their porosity, which aids in moisture absorption and contributes to the final aroma. Stainless steel and plastic containers were used sometimes in commercial settings that required larger batch sizes. The incubation period varied from 12 to 24 hours, depending on ambient temperature, package material, and the potency of the starter culture. Previous batch dahi was used as starter culture (back slopping). During winter fermentation need longer periods or additional insulation (wrapping pots in cloth), whereas summer temperatures accelerated coagulation. Starter in milk ratios also varied, resulting in differences in acidity, firmness, and sweetness between producers. During the fermentation process, charcoal was kept burning inside the bamboo basket to provide mild, continuous heat. This practice serves as a traditional temperature-control method, allowing the curd to ferment properly even when the ambient temperature changes. The technique appears to function as a natural incubator, promoting consistent texture and flavour development.

Storage and shelf life:

The storage conditions varied from short-term in room-temperature storage for household consumption to refrigeration for commercial sale. Producers selling in local markets typically kept *Mishti doi* for 1-3 days,

ARTICLE IN PRESS

depending on refrigeration availability. Traditionally, clay pots were covered with white paper. Modern packaging practices were limited, with most producers using traditional clay pot (common) or plain plastic containers.

Hygiene and safety observation:

Direct observations revealed significant variation in hygiene practices between sites. Well maintained production areas demonstrated routine utensil cleaning, separate preparation zones, and hand washing practices. In contrast, some households lacked systematic sanitation, as evidenced by open drainage, insufficient utensil cleaning, and poor environmental control. Producers frequently expressed confidence in the safety of *Mishti doi*, citing the lengthy boiling process and the acidic environment created during fermentation. However, only a small percentage demonstrated formal knowledge of microbial risks or standard food safety protocols. Spoilage incidents, such as souring and fungal growth, were more common during the humid months.

Mishti doi are produced in different part of West Bengal and production are varies greatly depending upon the local community and also different name was given by the local people. These are *Lal doi*, *Jhuri pata doi* and *Khir doi* [10, 11] Figure 2 and Figure 4. Many participants described the existence of several types of sweet curd prepared across different areas such as the dense (*Kheer doi*) and caramel-flavoured (*Lal doi*) and the lighter and moderately sweet *doi* popular in Bengal. The rich and creamy *Mishti doi* often sold in earthen pots, and locally adapted versions found in neighbouring districts where sweetness, texture, and fermentation methods vary. These regional subdivisions helped the study capture how preparation techniques, cultural practices, and local preferences contribute to unique styles of *Mishti doi* across different places.

ARTICLE IN PRESS

Lal doi

Lal doi was originated from Nabadwip (23.4037°N, 88.3659°E), located in the Nadia district of West Bengal. According to the book "Chaitanya Charitamrita," Sri Chaitanya Mahaprabhu loved this sweet. On Gaura Purnima, the birth anniversary of Sri Chaitanya Mahaprabhu (18 February 1486), Vaishnava worshippers fast all day till moonrise then eat "*Lal doi*" as an un-editable part of their meal [15]. It is also known as "*chakku doi*". Chakku means knife and the stickiness of *doi* is measured by inserting a knife (chakku) into the sweet so it is named as *chakku doi*. The basic ingredients are whole milk and sugar (4:3 v/v ratio). Local people used cow or buffalo milk for making this *doi*. The milk was simmered gently for 7 to 8 hours until its volume was reduced to one-third, developing a rich reddish hue (known as *lal*) from the prolonged cooking. With constant stirring, jaggery was added gradually to enhance the flavour. Once the desired sweetness was achieved, the mixture was allowed to cool to room

ARTICLE IN PRESS

temperature. A small amount of previously prepared *lal doi* (fermented curd) was added as a starter culture. The mixture was then carefully poured into traditional earthen pots (*matir bhar*) and placed near burning charcoal embers, covered with jute sacks to retain warmth. This environment allowed the curd to set naturally over 2 to 3 days. Various sized of earthen pot were (*matir bhar*), chosen to meet market demand.

Kheer doi

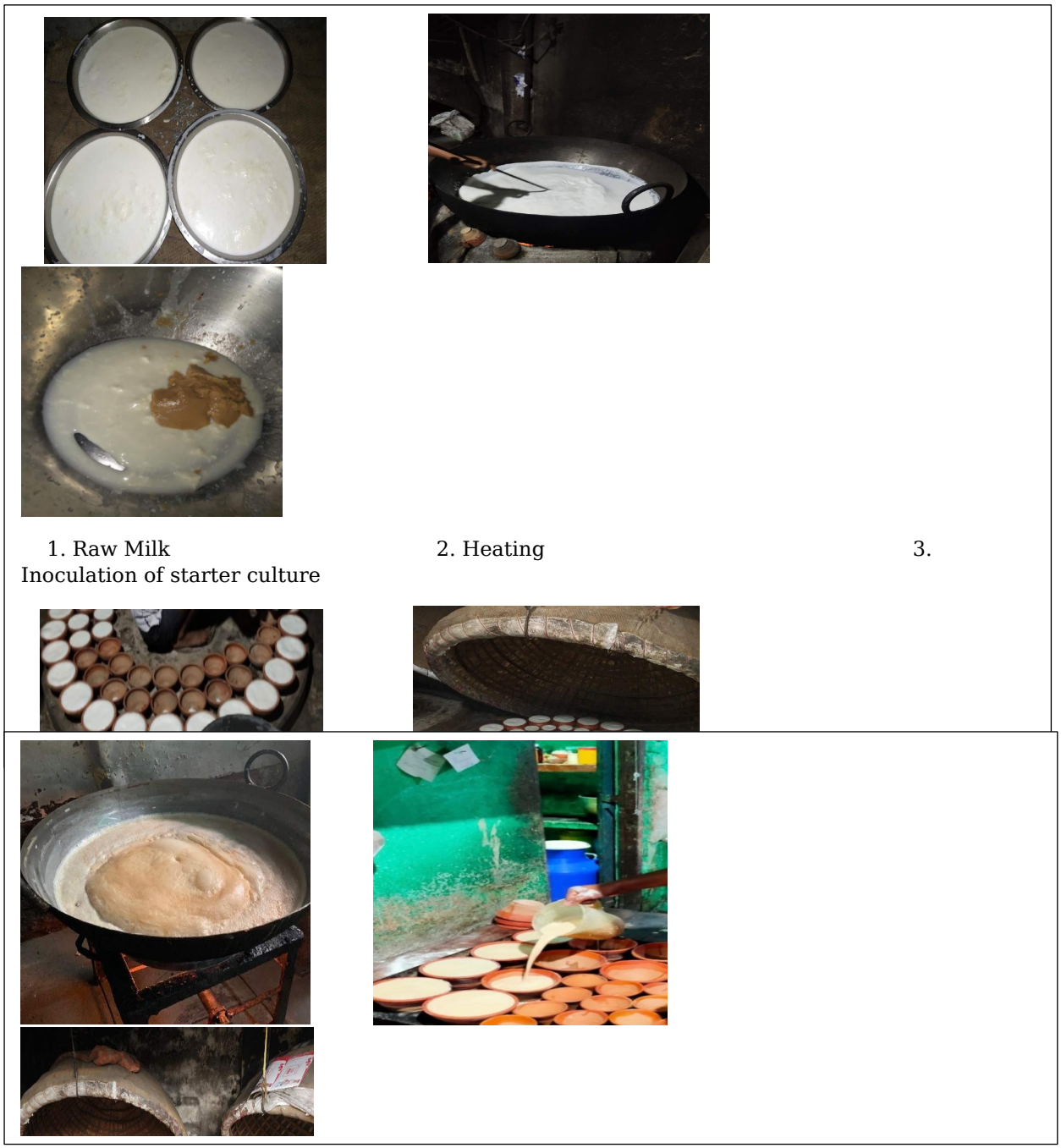
Gangarampur is situated in the Dakshin Dinajpur district of West Bengal (25.4009°N, 88.5324°E). Gangarampur is popular for *kheer* based dairy product. *Kheer*, is a popular South Asian dessert, has a rich and ancient history dating back more than 2,000 years. It was first mentioned in Ayurvedic texts and temple records of ancient India [16]. *Kheer* is usually made by boiling milk for prolonged time with sugar or jiggery. The slow cooking process helps the milk reduce to give thick and creamy texture. *Kheer* influenced similar dishes in Central Asia, the Middle East, and Southeast Asia, including *firni*, *rice pudding*, and *payesh*. Despite its numerous regional variations, *kheer* is still a beloved part of festivals and everyday meals, and it is one of the oldest and most enduring desserts in Indian cuisine. *Kheer* is usually made by boiling milk, sugar, or jiggery. Often flavoured with aromatic spices like cardamom and saffron, and garnished with nuts like almonds and cashews. In Gangarampur *kheer Doi*, is a type of traditional sweetened curd. The main ingredients are cow's milk and sugar. Milk was heated until "*Kheer*" has been reached.Then the caramelised sugar was mixed with "*kheer*" and previously prepared *dahi* was introduced to it. The bowl was kept in a warm environment for 8-10 hours. Final product was refrigerated for 2-3 hours to achieve the desired texture.

ARTICLE IN PRESS

Jhuri pata doi

It is a special kind of sweetened *dahi* that originated and found in Murshidabad (24.1759°N and 88.2802°E) district of West Bengal. Murshid Quli Khan (1717–27), the Diwan (chief revenue officer) of Bengal, under Aurangzeb, was fond of this sweet [15]. Usually, *dahi* is settled in earthen pots, but in this case, "*jhuri*" (bamboo basket) is used, so it is named as *Jhuri pata doi.* Milk was heated for 1-2 hours and then caramelised sugar was mixed with it. Inside this "*jhuri*", dried and condensed milk was used to seal the wholes of the basket. After coating of the basket, milk was poured to the *jhuri* and previously prepared *doi* was added for curding. The curd is thicker and yellowish in colour. It is less sweet than the other sweetened dahi.

Figure 3: Different types of *Mishti doi* (a) *Lal doi* or *Chakku doi* (b) *Kheer doi* (c) *Jhuri pata doi*



ARTICLE IN PRESS

Figure 4: (a) Preparation of *mishti doi* by using sugar (b) Preparation of *mishti doi* by using jaggery (*khejur gur*)

Socio-cultural significance of *Mishti doi*:

Dessert: *Mishti doi* is primarily consumed as a dessert in Bengal and the surrounding regions. It is a popular sweet treat, often served at the end of meals or on special occasions such as festivals and celebrations. Its rich, sweet, and slightly tangy flavor makes it a satisfying end to any meal. Traditionally served in a small clay pots, it offers a unique texture and taste experience that combines creamy smoothness with a hint of caramel. *Mishti doi* is sometimes paired with other sweets like *rasgulla* or garnished with some dry fruits. Its versatility has also inspired modern dessert innovations, such as *mishti doi* mousse, parfaits and fusion ice creams, allowing this traditional treat to find a place in contemporary menus as well.

Religious Offerings and Festive Gift: *Mishti doi* is more than just a dessert in Bengali culture it carries religious and spiritual meaning. It is often used as an offering in Hindu rituals and prayers, symbolizing purity, devotion, and good fortune. Made with milk and jaggery, ingredients considered sacred. *Mishti* doi is commonly presented to deities during pujas and festivals like Durga Puja, Lakshmi Puja, and Saraswati Puja. This product has been regarded sacred and is linked to traditional rituals such as marriage ceremonies, puja (worship), *Ashirbad* (blessings), *Upanayana*, and *Annaprasan* (the first rice-eating ceremony), where it is served as a sign of blessings and prosperity. The use of earthen pots to set *mishti doi* adds to its spiritual essence, reflecting simplicity and a deep connection to nature and tradition. *Mishti doi* is a traditional Bengali gift given during festivals, family gatherings, and visits to relatives which signifies prosperity and happiness.

ARTICLE IN PRESS

Culinary Ingredient: *Mishti doi* is an ingredient used in a variety of Bengali sweets and fusion desserts, including cakes, shakes (*doi-er-ghol*), and ice cream. Its thick, creamy texture and caramelised flavour make it an ideal base for innovative dessert recipes. Combining *mishti doi* with cake is a delightful fusion and it is a delicious way to blend classic Bengali flavors with a modern dessert format. This *Mishti Doi* Cake recipe yields a soft, moist treat, rich with the distinct caramel-like essence. *Doi-er-ghol* is a traditional Bengali yogurt-based drink that's refreshing, light, and perfect for hot days. It's similar to lassi but typically more diluted and can be made sweet or salty. For making *doi er ghol*, yogurt is blended until it becomes smooth and creamy, without lumps. Gradually cold water is added to get a thin, frothy, drinkable consistency not as thick as a *lassi*. For sweet *ghol* sugar and cardamom powder was added and for salty *ghol* salt and a pinch of roasted cumin powder was added. *Mishti do*i Ice cream fusion, blends the sweet, slightly tangy flavor of traditional mishti doi with the smooth, creamy texture of ice cream. *Mishti doi* is a popular Bengali dessert made by boiling full-fat milk, adding sweetness (usually with date palm jaggery or sugar), and letting it ferment with live cultures in clay pots. This process

ARTICLE IN PRESS

creates a thick, creamy yogurt with a caramel-like flavor and aroma. Ice cream, made from milk, cream, sugar, and stabilizers, is a perfect base for mixing in local and traditional tastes. Its rich, smooth texture allows it an excellent match for *mishti doi*.

Health benefits of *dahi*

Milk is a rich source of essential nutrients such as calcium, protein, vitamin D, phosphorus, riboflavin and vitamin A. It also contains important minerals like magnesium, zinc and iodine. Fermentation of milk leads to a number of organoleptic and nutritional changes, which are greatly influenced by the metabolic activity of starter cultures. *Dahi*, a popular traditional fermented milk product in India, differs significantly from its commercial counterparts due to variations in household production practices and the use of undefined microbial consortia as inoculants. Furthermore, the nutritional and technological value of traditional fermented milk products is influenced by a number of factors, including the composition of the milk, the microbial consortia present in the inoculum, and the unique methods of processing [18, 19, 20]. In rural North Bengal, people often believe that *Mishti doi* is not only nutritious but also beneficial to overall health. Many villagers claim that it aids digestion and keeps the stomach light, while also acting as a cooling food that reduces acidity and provides relief during hot weather. They consider it a natural source of strength because it is made from thick milk and it is frequently given to children to promote growth and nourishment. Some families also claim that *Mishti doi* is gentle on the stomach and ideal for those recovering from illness. In addition to these benefits, several local residents believe that eating *Mishti doi* on a regular basis improves skin appearance, making it look fresh and glowing because of its milk based nutrients. Overall, local communities regard *Mishti doi* as a nutritious, refreshing and health promoting food. For many years, *Doi* or *Dahi* (curd) is well-known for its health benefits (Figure 5). Fermented foods have been associated with improved digestion and overall wellness. During the preparation of *Dahi*, jaggery was used as sweetener and flavouring agent. Jaggery offers a better nutritional profile than the refined sugar due to its natural processing method. It is a good source of minerals like iron, calcium and magnesium. Proteins and carbohydrates undergo biotransformation through the milk fermentation process, by microorganisms or enzymes produced by them. Furthermore, indigenous enzymes contribute in the hydrolysis of milk protein and carbohydrate sources, which is responsible for the texture and flavour variations observed in the finished product. Differences in protein and carbohydrate content cause variations in body and texture scores among *dahi*. If the total solids content of a milk sample is low, the texture of *dahi* will be similarly low. The addition of fruit juice, date palm, and sugar to *dahi* or yoghurt improved its consistency [21]. "*Doi*" contains high levels of calcium (83 mg/100 g) and phosphorus (38 mg/100 g), which prevents arthritis and strengthens teeth and bones. These foods reduce cortisol secretion (an imbalanced cortisol level causes more fat accumulation in the body) and aid in weight loss. As a probiotic food (rich in *Lactobacillus* and

ARTICLE IN PRESS

ARTICLE IN PRESS

Leuconostoc sp.), it boosts immunity and aids the proper functioning of the digestive system [16]. The scientific community's interest in studying the therapeutic properties of food products was influenced by Ellie Metchnikoff's seminal finding linking about consumption of traditional fermented milk products linking to the longevity of Bulgarian peoples [24]. The use of fermented dairy products like *dahi* in traditional ayurvedic formulations, such as *Panchgavya* dating back to 600 AD, reinforces their status as functional foods. *Dahi* is recommended for preventing digestive disorders including constipation, diarrhoea and dysentery and for limiting the growth of intestinal bacteria. *Dahi* has also been reported to reduce blood cholesterol levels.

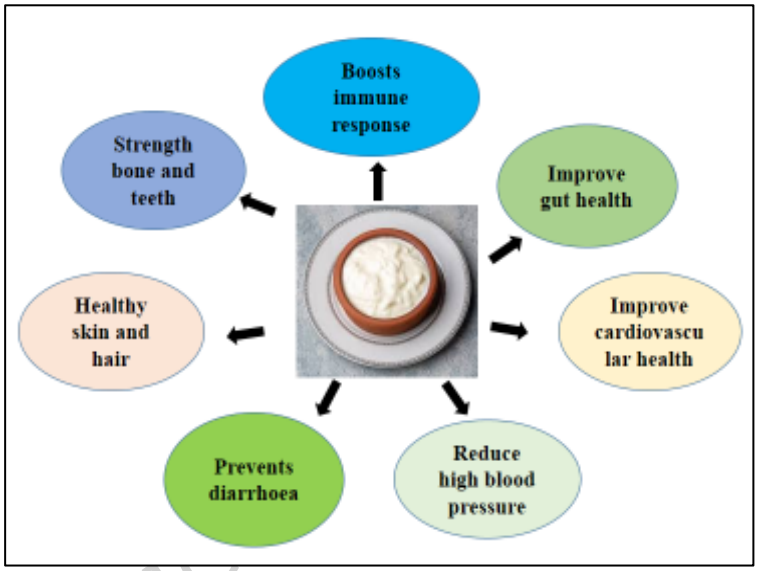


Figure 5: Health benefits of Dahi

However, little is known about specific health benefits *Mishti doi*. Unlike commercial dahi, *Mishti doi* is made with different ingredients, heating processes, and fermentation conditions, which may contribute to its distinct nutritional and functional properties. The use of caramelised sugar or jaggery, extended milk boiling, and earthen-pot fermentation can all create a unique biochemical environment that may support the growth of different or more diverse microorganisms than commercially produced dahi. This variation in ingredients and preparation techniques suggests that *Mishti doi* may contain a microbial community with potentially beneficial properties not found in industrial products. Variations in sugar type, milk concentration and natural starter cultures may have an additional impact on microbial composition, metabolic activity, and bioactive compound formation. These differences highlight the possibility that traditional *Mishti doi* may provide additional health benefits beyond those of standard commercial curd, emphasising the importance of scientifically documenting, analysing, and comparing the microbial

ARTICLE IN PRESS

profiles and functional properties associated with this traditional product. Metagenomic and metagenome-assembled genome (MAG) tools can be used to identify biomarkers for genes responsible for various health-promoting functions present in *Mishti doi*. Different types of *dahi* are often made using a mixed culture of *Streptococcus lactis*, *Streptococcus thermophilus, Lactobacillus bulgaricus, and Lactobacillus plantarum*. Together with culture, 10 to 12% sugar is added for making sweet *dahi* [25]. Considering the existing reports on several health-promoting properties as well as their generally recognized as safe (GRAS) status of LAB, they can be widely used in the developing of new fermented milk products**.** *Lactococcus lactis lactis*, *Lactococcus lactis sp*. *Cremoris*, *Lactobacillus kefir*, *Lactobacillus kefiranofaciens*, *Lb. acidophilus*, *S. thermophilus*, *Lb. delbrueckii ssp*. *Bulgaricus*, *Lactobacillus delbrueckii ssp. lactis*, *Lb. helveticus*, *Lb. casei*, are the functional dairy starter cultures [26]. The metabolic process of various beneficial microbes, particularly the LAB, has a direct impact upon the alterations in biochemical components in the milk leads to changing the rheological and functional properties, resulting nutritional enhancement of the final product. LAB produces metabolites include bacteriocins, enzymes, organic acids, gases, hydrogen peroxide, vitamins, exopolysaccharides, diacetyl, acetoin, and acetaldehyde, low molecular weight compounds such as reuterin and phenyl lactic acid, antifungal metabolites (cyclic dipeptides and phenyllactic acid), proteinaceous compounds, 3-hydroxylated fatty acid, and so on [8]. Scientific and industrial interests in creating foodstuffs in addition to delivering nutritional value, can also improve the consumer's general health [27, 28]. So probiotic dairy products are therefore of great interest to both consumers and researchers who are looking for nutritious foods with enhanced health advantages such as enhancing the bioavailability of vitamins and minerals, protecting against infectious agents, immunomodulatory effects, anti-allergenic effects, anti-obesity effects, and anti-anxiety effects, decrease the symptoms of lactose intolerance, among others [29, 30, 31, 32]. Probiotic is the live microorganisms that can confer at least one health benefit when given in sufficient amount to host. Probiotics have grown in significance in both basic and applied research and also a valuable economic topic of interest as a result of their rising popularity [33]. *Bifidobacterium* (*adolescentis, casei, fermeanimalis, bifidum, breve, and longum*) and *Lactobacillus* (*acidophilus, casei, fermentum, gasseri, johnsonii, paracasei, plantarum, rhamnosus, and salivarius*) was well-studied species, that "promotes a healthy gut flora" and having general health advantages notably for the gastrointestinal tract and immune function [34, 35]. Consuming probiotics can promote intestinal health by regulating the microbiota, stimulate the immune system's growth and function, increase the bioavailability of nutrients, and lower the chance of developing a number of diseases [36] mentioned in Table 2. LAB-driven fermentations frequently produce by-products that are bioactive and have a variety of health-promoting effects.

ARTICLE IN PRESS

Indigenous people believe *Mishti doi* improves digestion, strengthens the body and promotes overall health. These beliefs could be linked to the presence of beneficial microorganisms that develop naturally during traditional fermentation. If the bacterial strains used as the starter culture in traditional *Mishti doi* are scientifically identified and confirmed as potential probiotics, it will clarify the health related perceptions commonly held by local communities. Furthermore, confirming probiotic properties would highlight the functional value of traditional fermentation practices, implying that *Mishti doi* may have more health benefits than commercial *Dahi* made with standardised cultures. The link between microbial composition and perceived health effects emphasises the importance of characterising native starter cultures and assessing their probiotic potential. Industrialisation and modernisation of traditional *Mishti doi* production presents a number of challenges. The preparation process varies greatly across regions and individual households, making it difficult to standardise fermentation steps, starter cultures, ingredient proportions, and heating methods. Traditional producers rely on experience based methods, indigenous starter cultures with unknown microbial compositions make difficulty to ensure consistent product quality and safety on a large scale. Furthermore, the high sugar content, caramelisation of milk and distinct regional sensory characteristics present in the products, face challenges when adapted to large-scale automated systems. Maintaining the authentic taste while meeting modern hygiene, shelf life, packaging and cold-chain distribution standards remains a significant challenge.

ARTICLE IN PRESS

Table 2: Functional properties of Dahi:

Health benefits	**Name of the Bacteria**	**Method of analysis**	**References**
Inhibitory activity against *P. aeruginosa* and *S. aureus*	*Lactiplantibacillus plantarum* ADMH 97 *Lactiplantibacillus pentosus* ADMH 73 *Lacticasebacillus rhamnosus* ADMH 2 *Lactobacillus pontis* ADMT 16 *L. plantarum* ADMG 1	*in-vitro*	[37]
The DPPH radical scavenging activity (with percentage of 75% and 44.85%).	*L. plantarum* ADMH 97 *L. plantarum* ADMG	*in-vitro*	[37]

ARTICLE IN PRESS

Serum pro-inflammatory (TNFα, IFNγ), and anti-inflammatory (IL-10) cytokine levels were increased. Lymphocyte counts were increase from 3707±1551 to 6312±1937 after 6 weeks of treatment.	*Lactobacillus bulgaricus* and *Streptococcus thermophiles*	Clinical trial (children patients suffering from protein energy malnutrition)	[38]
Free amino acids and γ-Aminobutyric acid (GABA), polyphenols availability, antioxidant activity (up to 54%)	*L. rhamnosus* SP1, *W. confusa* DSM 20194, *L. plantarum* T6B10	*in-vitro*	[39]
[alpha]- amylase inhibitory activity associated with the management of postprandial hyperglycemia	*Lactobacillus (Lb.) plantarum*	*in-vitro*	[40]
Cholesterol-lowering activity, EPS production, antioxidant property	*L. acidophilus* (WFA1, WFA2, and WFA3)	*in-vitro*	[41]
EPS production, antibacterial activity against pathogenic strains *E. coli* ATCC25922, *S. aureus* ATCC6538, *P. aeruginosa* ATCC25923, and *L. monocytogenes*	*S. thermophiles*	*in-vitro*	[42]
Increasing immune cell levels in malnourished children, especially in the age group of 7–8. Monocytes (3.11 to 5.21), neutrophils (26 to 38), basophils (0.72 to 0.84), and lymphocytes (27 to 35)	*Lactobacillus acidophilus*, LBKV-3	Clinical trial	[43]

ARTICLE IN PRESS

ARTICLE IN PRESS

Reduced the adherence of foodborne pathogen *Listeria monocytogenes* AMD K2 (47.46 ± 0.17 %) to Caco-2 cells.	*Lactococcus lactis* AMD17	*in-vitro*	[44]
Weight gain in albino rat (129.33 g) and the serum cholesterol level of these rats decreased gradually, with a significant reduction of 4.1 mg/100 ml	*Bifidobacterium bifidum* and *Lactobacillus acidophilus*	*in-vivo*	[45]
Exhibit hypocholesteromic effects and probiotic attributes	*Lactobacillus plantarum YD5S, L. plantarum YD9S, L. pentosus YD8S, L. paraplantarum YD11S*	*in-vitro*	[31]

Limitation of the study

Despite careful planning, study has some limitation. The study included a wide variety of *Mishti doi* producers and consumers, representing various age groups, educational levels, and socioeconomic backgrounds. The majority of traditional producers worked for small family-owned dairy farms, with 90 % reporting that their preparation knowledge had been passed down through two or more generations. Formal training in food processing was scarce, with the majority relying on inherited practices and experiential learning. The sample coverage was limited, as only a subset of local producers and households in North Bengal could be visited within the time and resources available. This means that the findings may not be representative of all traditional *Mishti doi* practices in the wider region.

The study relied entirely on field data and descriptive responses, with no controlled fermentation experiments conducted in a laboratory setting. As a result, scientific validation of fermentation conditions such as temperature, microbial population, and precise fermentation time could not be determined. Seasonal effects on fermentation could not be assessed. Furthermore, the study primarily focused on small scale traditional producers, and commercial dairy practices were excluded.

Future prospects

The ethnic origins of traditional fermented milk products, along with limited research into their technological and functional properties, have kept these products largely confined to their regions of origin. However, recent advancements in dairy and food science have revealed the vast

ARTICLE IN PRESS

ARTICLE IN PRESS

market potential of these indigenous fermented milk products. As a result, many food companies are now working to bring attention to this emerging segment, aiming to make these traditional products more popularise. Global awareness of the health benefits of bioactive dietary compounds with specific functions has rapidly increased. The nutraceutical market was valued at $454.55 billion, with a projected compound annual growth rate (CAGR) of 9% from 2021 to 2033 [46, 47]. The rising demand for functional foods and natural dietary supplements has driven food researchers to pay greater attention to fermented food products [48]. Modern fermentation processes use selected starter cultures with desirable traits to ensure consistency and commercial success. By integrating data-driven approaches such as culturomics, metagenomics, and metabolomics, research on these fermented milk products can enhance culturing methods of microorganisms, which are traditionally involved in fermentation. This approach also helps validate traditional beliefs about the benefits of these products. These findings could regulate the guidelines for production technology and contribute to the development of a new segment of functional foods in India.

Conclusion

Fermented milk products is a part of the human diet for centuries due to their pleasant taste and also by their cultural variations. *Mishti doi*, a traditional fermented milk product of Bengal and the fermentation process is associated with local culture and varies greatly from one place to another. Indo-Aryan ethnoliguistic group are known as bangalees or Bengalis, are historically connected to the production of *Mishti doi*. Local people generate income by selling their creations. This survey provides a basic overview of the traditional methods, local perceptions, and cultural significance of *Mishti doi* preparation in selected regions. The findings demonstrate that producers rely on long established, experience based techniques, often employing indigenous starter cultures with microbial characteristics that have yet to be scientifically investigated. Growing consumer awareness has influenced the research into the functional aspects of indigenous fermented milk products, with an aim to explore their potential as biotherapeutic and nutraceutical alternatives in medicine. To ensure the safety of *Mishti doi* for human consumption, standardized protocols, fermentation process guidelines, and quality control measures must be established for its production. Overall, this survey provides the groundwork for future research aimed at characterising the microbial profile, evaluating probiotic properties and providing scientific evidence for the health benefits traditionally associated with *Mishti doi*.

Declarations

Ethics approval

ARTICLE IN PRESS

ARTICLE IN PRESS

The study was conducted in accordance with the ethical principles of the Declaration of Helsinki and followed relevant national and institutional guidelines. The survey was conducted by open ended questionnaire and face to face interviews, it did not involve any kind of clinical procedure or risk to the respondents of North Bengal region. Hence, institutional ethics committee of the University of North Bengal waived the need for approval.

Consent to participate

Informed consent was obtained from all participants prior to data collection. Participants were clearly informed about the purpose of the study, any potential risks and benefits involved. All participants interviewed were above 18 years and have agreed voluntary to give information on the ethnic heritage of their region.

Clinical trial number

Not applicable

Consent to publish

Not applicable

Data Availability

All data and materials have been presented in this paper.

Competing interest

The authors have no competing interest to declare.

Funding

Not applicable.

Authors' Contributions

PC has collecting the data and written the manuscript, and KG has revised and finalised the manuscript.

Acknowledgements

The author would like to express sincere gratitude to the Department of Food Technology, University of North Bengal.

Authors' information

Priya Chakraborty: Research Scholar, [1]Food Microbiology laboratory, Department of Food Technology, University of North Bengal, Raja Rammohunpur, Siliguri, West Bengal, 734013, India, rs_priya@nbu.ac.in

ARTICLE IN PRESS

ARTICLE IN PRESS

Corresponding author: *Dr. Kriti Ghatani, Assistant Professor, [1]Food Microbiology laboratory, Department of Food Technology, University of North Bengal, Raja Rammohunpur, Siliguri, West Bengal, 734013, India, ghatanik@nbu.ac.in

References:

1. West Bengal Population, Census 2021.
2. Ghosh, S. (2016). Colonial Economy in North Bengal, 1833-1933. Paschimbanga Anchalik Itihas O Loksanskriti Charcha Kendra.
3. Rawat K, Kumari A, Kumar S, Kumar R, Gehlot R. Traditional fermented products of India. Int J Curr Microbiol App Sci. 2018;7(4):1873-83.
4. Ghosh K, Mondal SP, Mondal KC. Ethnic fermented foods and beverages of West Bengal and Odisha. Ethnic Fermented Foods and Beverages of India: Science History and Culture. 2020:647-85.
5. Tamang JP, Samuel D. Dietary cultures and antiquity of fermented foods and beverages. Fermented foods and beverages of the world. 2010 Jul 1;1(1.3):1.
6. Sarkar P, Dh LK, Dhumal C, Panigrahi SS, Choudhary R. Traditional and ayurvedic foods of Indian origin. Journal of Ethnic Foods. 2015 Sep 1;2(3):97-109.
7. Ghatani K, Thapa S, Chakraborty P. Hypolipidemic and hypoglycemic nature of lactobacillus strains in fermented vegetable and dairy products. Frontiers in Food Science and Technology. 2022 Oct 20;2:955990.
8. Mallappa RH, Balasubramaniam C, Nataraj BH, Ramesh C, Kadyan S, Pradhan D, Muniyappa SK, Grover S. Microbial diversity and functionality of traditional fermented milk products of India: Current scenario and future perspectives. International Dairy Journal. 2021 Mar 1;114:104941.
9. Saikia G, Mishra BK. Growth of Lactic Acid Bacteria in Milk for the Preparation of Functional Frozen Misti Dahi (Sweet Curd). Int J Sci Res Sci Technol. 2017;3(8):22-6.
10. Ghosh J, Rajorhia GS. Selection of starter culture for production of indigenous fermented milk product (Misti dahi). Le Lait. 1990;70(2):147-54.
11. Tamang JP, Thapa N, Bhalla TC, Savitri. Ethnic fermented foods and beverages of India. Ethnic fermented foods and alcoholic beverages of Asia. 2016:17-72.
12. Josephsen J, Jespersen L. Starter cultures and fermented products. InHandbook of food and beverage fermentation technology 2004 Mar 19 (pp. 48-80). CRC Press.
13. Agarwal KN, Bhasin SK. Feasibility studies to control acute diarrhoea in children by feeding fermented milk preparations

ARTICLE IN PRESS

Actimel and Indian Dahi. European Journal of Clinical Nutrition. 2002 Dec;56(4):S56-9.
14. Shangpliang HN, Tamang JP. Metagenome-assembled genomes for biomarkers of bio-functionalities in Laal dahi, an Indian ethnic fermented milk product. International Journal of Food Microbiology. 2023 Oct 2;402:110300.
15. Sarkar T, Salauddin M, Paul A, Choudhury T, Chakraborty R, Imran F. The essence of Bengal's ethnic sweetmeats: an exploratory journey through history, tradition, and culture. Journal of Food Quality. 2023;2023(1):5008420.
16. Sarkar S, Sakar M, Majhi R, Chatterjee K, Sikder B, Sur A. Process standardization for enhancing bio functionality of Dahi. J. Nutr. Health Food Eng. 2019;9:1-6.
17. Dey I. Who owns Bogurar Doi? An ethnography of placemaking and craftsmanship in Bangladesh. Gastronomica: The Journal of Food and Culture. 2022 Aug 1;22(3):35-45.
18. Narender Raju P, Pal D. The physico-chemical, sensory, and textural properties of Misti Dahi prepared from reduced fat buffalo milk. Food and Bioprocess Technology. 2009 Mar;2:101-8.
19. Tamang JP, editor. Ethnic fermented foods and beverages of India: science history and culture. Springer Nature; 2020 Mar 2.
20. Tigga A, Mallappa RH, Muniyappa SK, Kadyan S, Pradhan D, Niharika ES, Grover S. 16S metagenomics and metabolomics unveil the microbial compositions and metabolite profiles in Dahi, a traditional Indian fermented milk product prepared by the backslopping method. Journal of Food Science and Technology. 2024 Aug 22:1-4.
21. Hossain AA, Habib R, Islam MN, Rakib MR. Use of date-palm (Phoenix dactylifera L.) brown sugar as a sweetener for making Dahi. Bangladesh Journal of Animal Science. 2014 Jun 30;43(1):62-7.
22. Afrin S, Habib MR, Islam MA, Harun-ur-Rashid M. Physical, chemical and microbiological qualities of dahi collected from Bogra and Mymensingh district of Bangladesh. Asian-Australasian Journal of Bioscience and Biotechnology. 2016 Apr 30;1(1):134-40.
23. Matin A, Rahman N, Islam T, Ahmed FB. Effect of adding coconut milk on the physicochemical, proximate, microbial and sensory attributes of «Dahi». Ukrainian Journal of Food Science. 2020 Jun 30;8(1):49-57.
24. Ozen M, Dinleyici EC. The history of probiotics: the untold story. Beneficial microbes. 2015 Mar 22;6(2):159-66.
25. Akter MN, Hashim R, Sutriana A, Siti Azizah MN, Asaduzzaman M. Effect of Lactobacillus acidophilussupplementation on growth performances, digestive enzyme activities and gut histomorphology of striped catfish (Pangasianodon hypophthalmusSauvage, 1878) juveniles. Aquaculture Research. 2019 Mar;50(3):786-97.
26. Ayivi RD, Gyawali R, Krastanov A, Aljaloud SO, Worku M, Tahergorabi R, Silva RC, Ibrahim SA. Lactic acid bacteria: Food

ARTICLE IN PRESS

safety and human health applications. Dairy. 2020 Oct 29;1(3):202-32.
27. da Silva TF, Glória RD, Americo MF, Freitas AD, de Jesus LC, Barroso FA, Laguna JG, Coelho-Rocha ND, Tavares LM, Le Loir Y, Jan G. Unlocking the potential of probiotics: A comprehensive review on research, production, and regulation of probiotics. Probiotics and Antimicrobial Proteins. 2024 Oct;16(5):1687-723.
28. Zaib S, Hayat A, Khan I. Probiotics and their beneficial health effects. Mini reviews in medicinal chemistry. 2024 Jan 1;24(1):110-25.
29. Linares DM, Gómez C, Renes E, Fresno JM, Tornadijo ME, Ross RP, Stanton C. Lactic acid bacteria and bifidobacteria with potential to design natural biofunctional health-promoting dairy foods. Frontiers in microbiology. 2017 May 18;8:846.
30. Kumari M, Patel HK, Kokkiligadda A, Bhushan B, Tomar SK. Characterization of probiotic lactobacilli and development of fermented soymilk with improved technological properties. LWT. 2022 Jan 15;154:112827.
31. Ghatani K, Tamang B. Assessment of probiotic characteristics of lactic acid bacteria isolated from fermented yak milk products of Sikkim, India: Chhurpi, Shyow, and Khachu. Food Biotechnology. 2017 Jul 3;31(3):210-32.
32. Maftei NM, Raileanu CR, Balta AA, Ambrose L, Boev M, Marin DB, Lisa EL. The potential impact of probiotics on human health: An update on their health-promoting properties. Microorganisms. 2024 Jan 23;12(2):234.
33. Sakandar HA, Zhang H. Trends in Probiotic (s)-Fermented milks and their in vivo functionality: A review. Trends in Food Science & Technology. 2021 Apr 1;110:55-65.
34. Zielinska D, Kolozyn-Krajewska D. Food-origin lactic acid bacteria may exhibit probiotic properties. Biomed. Res. Int.. 2018 Oct;2018:5063185.
35. Han J, McClements DJ, Liu X, Liu F. Oral delivery of probiotics using single-cell encapsulation. Comprehensive Reviews in Food Science and Food Safety. 2024 May;23(3):e13322.
36. Nagpal R, Kumar A, Kumar M, Behare PV, Jain S, Yadav H. Probiotics, their health benefits and applications for developing healthier foods: a review. FEMS microbiology letters. 2012 Sep 1;334(1):1-5.
37. Muhammed Salman PA, Nishida NP, Krishnendhu KU, Rahila MP, Divya KB, Beena AK, Rejessh R. Antimicrobial and antioxidant potential of indigenous lactic acid bacteria from raw goat milk and dahi.
38. Archana, K., Pooja, D., & Sunil, G. (2022). Dahi (an immunonutrient) in India Culture. *International Journal of Medical Science and Clinical Research Studies*, *2*(6), 505-511.
39. Lorusso A, Coda R, Montemurro M, Rizzello CG. Use of selected lactic acid bacteria and quinoa flour for manufacturing novel yogurt-like beverages. Foods. 2018 Apr 1;7(4):51.

ARTICLE IN PRESS

40. Tirwa RK, Tamang B, Chaurasia LK. Lactobacillus plantarum Isolates from Homemade Dahi as a Potential Probiotic with. management. 2018;12:1167-80.
41. Farid W, Masud T, Sohail A, Ahmad N, Naqvi SS, Khan S, Ali A, Khalifa SA, Hussain A, Ali S, Saghir M. Gastrointestinal transit tolerance, cell surface hydrophobicity, and functional attributes of Lactobacillus acidophilus strains isolated from Indigenous Dahi. Food Science & Nutrition. 2021 Sep;9(9):5092-102.
42. Taj R, Masud T, Sohail A, Sammi S, Naz R, Sharma Khanal BK, Nawaz MA. In vitro screening of EPS-producing Streptococcus thermophilus strains for their probiotic potential from Dahi. Food science & nutrition. 2022 Jul;10(7):2347-59.
43. Hajare ST. Effects of potential probiotic strains LBKV-3 on Immune Cells responses in Malnutrite children: a double-blind, randomized, Controlled trial. Journal of Immunoassay and Immunochemistry. 2021 Sep 3;42(5):453-66.
44. Manhar AK, Saikia D, Borah A, Das AS, Gupta K, Roy R, Mahanta CL, Mukhopadhyay R, Mandal M. Assessment of goat milk-derived potential probiotic L. lactis AMD17 and its application for preparation of dahi using honey. Annals of Microbiology. 2016 Sep;66:1217-28.
45. Vijayendra SV, Gupta RC. Assessment of probiotic and sensory properties of dahi and yoghurt prepared using bulk freeze-dried cultures in buffalo milk. Annals of microbiology. 2012 Sep;62:939-47.
46. Chourasia R, Chiring Phukon L, Abedin MM, Padhi S, Singh SP, Rai AK. Bioactive peptides in fermented foods and their application: A critical review. Systems Microbiology and Biomanufacturing. 2023 Jan;3(1):88-109.
47. Functional Food and Beverage Market Size | Global Report, 2028. 2020. https:// www. fortu nebus iness insig hts. com/ funct ional- foods- market- 102269. Accessed 28 Jun 2022.
48. Dullius A, Rama GR, Giroldi M, Goettert MI, Lehn DN, de Souza CF. Bioactive peptide production in fermented foods. InCurrent Developments in Biotechnology and Bioengineering 2022 Jan 1 (pp. 47-72). Elsevier.